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S2SDeepArr: Sequence To Sequence Deep Learning Architecture for Arrhythmia Detection Under the Inter-patient Paradigm

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Abstract

Electrocardiogram (ECG) signal analysis is a crucial tool for enhancing the efficacy of clinical diagnosis, particularly in detecting arrhythmias. However, its performance tends to degrade under the inter-patient paradigm, especially for minority sample categories. To address this issue and enhance the detection performance of these less represented classes within the inter-patient test protocol, this paper proposes a novel hybrid framework. This framework integrates specialized blocks from convolutional networks, namely DeepArr CNN, with sequence-to-sequence BiLSTM models. The proposed approach involves extracting local ECG features from a sequence of heartbeats utilizing DeepArr CNN. These extracted feature maps are then fed into the RNN encoder-decoder to facilitate fusion with the feature maps of neighboring heartbeats. Our proposed method, which adhered to the AMII standard of the MIT-BIH arrhythmia database and operated within the inter-patient paradigm, achieved impressive accuracy rates. Specifically, it yielded accuracy rates of 99.92% and 99.81% for three and five class scenarios, respectively. The performance measures across different classes are as follows: for the N class, sensitivity (SEN) is 99.81%, positive predictive value (PPV) is 99.73%, and specificity (SPEC) is 97.81%; for the S class, SEN is 92.97%, PPV is 96.01%, and SPEC is 99.85%; for the V class, SEN is 99.97%, PPV is 99.26%, and SPEC is 99.95%; and for the F class, SEN is 97.42%, PPV is 95.70%, and SPEC is 99.97%. Even when dealing with an unbalanced dataset, our proposed solution consistently yields remarkable outcomes. Specifically, it achieves an accuracy rate of 99.24% in the three-class scenario and 98.75% in the five-class scenario. These experimental results underscore the effectiveness of our S2SDeepArr model for ECG classification under the inter-patient paradigm, as demonstrated across various experimental cases.

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1. Introduction

An Electrocardiogram (ECG) is the most informative tool to register heart activities and diagnose cardiovascular disease which is associated with arrhythmia. Automatic ECG signal analysis for arrhythmia classification can help cardiologists in making high-accuracy diagnoses in real-time and improving clinical efficiency.

Automatic arrhythmia classification has been a persistent challenge. Numerous machine learning algorithms have attained promising outcomes in arrhythmia detection through a combination of well-designed feature extraction processes and classifiers. The basic feature used for ECG classification is the distance between two R peaks of the ECG signal and the morphological features such as morphological distance, amplitude morphology, and area morphology [1]. Also, researchers have presented several techniques for ECG feature extraction that encompass higher-order statistics (HOS) [2], projection features [3], LBP, wavelet [2], and Fourier [4]. These techniques are based on statistical or physiological theories to represent the ECG signals. Numerous ECG classifiers, such as support vector machines (SVM) [2], decision trees, Linear discriminant analysis, K-Nearest Neighbor (KNN), Bayes network, and multi-layer perceptrons (MLP) [5] have been proposed by researchers and shown interesting results. Despite these traditional methods with handcraft features and noise removal techniques have demonstrated satisfactory performance, recently deep learning have achieved remarkable success in various domains ranging from image classification, computer vision, and natural language processing (NLP). Notably, these deep learning techniques have shown promising and applicable outcomes in the field of biomedical signal processing. Based on this achievement, researchers have developed deep learning algorithms to classify arrhythmia. Additionally, with the quick evolution of convolutional neural networks (CNNs) in image classification and pattern recognition, algorithms built with convolutional layers, batch normalization, and max or average pooling layers with linear layers at the end of the model have shown promising outcomes in arrhythmia classification. Moreover to CNN, recurrent neural networks (RNNs) have been used for time-series analysis and ECG signal classification [6]. Also, a combination of CNN and RNN have been applied in arrhythmia classification [7].

Two research paradigms were presented in the state-of-the-art which are the intra-patient and the inter-patient paradigms. For the first paradigm, the state of the art have reached the highest levels of performance but the second paradigm is still taken into consideration. Scientists have attempted diverse approaches to enhance this paradigm such as the SVM and the integrated SVMs methods which attained an accuracy of 93.10% [3] and 94.5% [2] respectively, also linear discriminant analysis (LDA) achieved 90.29% [5], and multilayer perceptron obtained 96.72% [8]. Moreover, effective deep learning architectures such as CNN, have attained 95.33% [9] and 91.44% [10]. Nevertheless, these achievements exhibit poorer performance in comparison to the intra-patient paradigm. Additionally, fewer categories' performance is unsatisfactory, frequently dropping below 50% or even worse.

Various efforts have been undertaken to enhance performance within the inter-patient paradigm. Acknowledging the power advantages of taking into account the relation between adjacent heartbeats, certain researchers have investigated the application of sequence-to-sequence models for performance improvement. Here some research used seq2seq under the inter-patient paradigm for arrhythmia detection. For example, Mousavi et al. [11] joined CNN with BiLSTM, which attained an important result with an accuracy of 99.53%. According to Mousavi's outcomes, many researchers [6, 12, 13, 14] built upon these findings, Li et al. [6] introduced a new combination of the Weight Capsule network and the seq2seq model achieving an accuracy of 99.85%. Also, Zhou [14] united a multi-layer perceptron and Weight Capsule (WCapsule) network with the seq2seq model, resulting in an updated accuracy of 99.83%. Meng et al. [13] proposed an additional preprocessing method, reaching a high overall accuracy of 99.32%. It may suggest that the challenge of inter-patient ECG classification in the MIT-BIH dataset has been effectively addressed.

Despite the notable success reached by the aforementioned deep learning methods in arrhythmia classification, there is still a need to enhance the deep learning model's performance in the case of imbalanced classes of ECG signals.

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Currently, the main techniques used by deep learning to deal with sample imbalance in arrhythmia classification are sampling and weighting. Earlier studies have employed attention mechanisms [15, 12] and two-level networks [12, 14, 16, 6] to enhance models' capacity for learning and capture more profound features of heartbeats. Nevertheless, certain approaches mentioned above are complex and complicated. The Synthetic Minority Over-Sampling TEchnique (SMOTE) [17] was used to create novel samples and augment the beat size. SMOTE showed the best results in dealing with imbalanced data problems for the intra-patient paradigm [18] and also in the inter-patient paradigm especially in the sequence-to-sequence method [11, 12] for arrhythmia classification.

As discussed earlier, the sequence-to-sequence deep learning method demonstrated superior performance compared to other techniques. In the feature extraction phase, Mousavi et al. [11] employed a convolutional layer to extract only local features, which proved insufficient for ECG signal analysis. They also incorporated a pooling layer, leading to overfitting issues, thereby causing the LSTM encoder-decoder to struggle with effective feature weighting. To address this limitation, other researchers [14, 12] have adopted the attention mechanism. For instance, some approaches combine a Transformer with CNN and a denoising autoencoder, while others utilize local channel-wise attention to weight morphological features and contextual Bahdanau's attention to weight semantic features captured by a seq2seq LSTM module. Despite showing promising results, these architectures are complex and necessitate a denoising step. In light of this analysis, we use our proposed DeepArr model [19], which effectively extracts ECG signal features without high complexity or the need for denoising methods such as autoencoders. DeepArr comprises two successive convolutional layers to capture both local and temporal features, along with a dropout layer to mitigate overfitting. Consequently, we have drawn inspiration from S2SDeepArr by [11, 19] to achieve robust performance under the inter-patient paradigm. According to experimental results, S2SDeepArr proficiently tackles the problem of class imbalance, showcasing a top-tier overall average accuracy (ACC) and exceptional performance across all categories. Compared to deep learning approaches and sequence-to-sequence methods, S2SDeepArr has attained a performance rate of 96% across all categories, the main contributions include:

- We introduce a new model, S2SDeepArr, for arrhythmia detection, utilizing an innovative sequence-to-sequence network architecture that combines CNN and BiLSTM encoder-decoder components. Our model processes raw ECG heartbeat data, assigning a feature map to each heartbeat and organizing consecutive heartbeats into a sequence of feature maps, which enhances the feature representation of individual heartbeats by allowing them to attend to their neighbors. Comprehensive experimental studies under diverse cases demonstrate that S2SDeepArr achieves satisfactory performance without relying on handcrafted features, even when working with imbalanced datasets.
- To guarantee the S2SDeepArr model performance, we tested our model under three-classes and five-classes scenarios, which obtained promising ACC, SEN, PPV, and SPEC in both scenarios under AMII standard with test protocol.

2. Materials

2.1. MIT-BIH dataset and division standard

We used the publicly accessible ECG database MIT-BIH Arrhythmia Database [20] to evaluate our method. The MIT-BIH Arrhythmia dataset comprises 48 electrocardiogram recordings, each record includes a dual-channel (MLII, V1/V2/V4/V5 according to the subject) of about half-hour of 47 subjects. Recordings were sampled into 360 samples for each channel. To evaluate our purposes for training and testing we used only the MLII lead. MITBIH dataset is recommended by the AAMI standard [21] which groups the MIT-BIH arrhythmia annotations into five arrhythmia categories: Normal beat (N), Supraventricular ectopic beats (S), Ventricular ectopic beats (V), Fusion of ventricular and normal beat (F), Unclassifiable beat (Q).

We must avoid including the same patient's heartbeats in the train and test sets at the same time because our research is predicated on the inter-patient paradigm. To compare our research with the state of the art, this study implemented the inter-patient paradigm introduced by Chazal et al. [22] that split the 47 subjects of the MIT-BIH database into two groups one for training and the other for testing:

- Training set DS1= {101, 106, 108, 109, 112, 114, 115, 116, 118, 119, 122, 124, 201, 203, 205, 207, 208, 209, 215, 220, 223, 230}
- Testing set DS2={100, 103, 105, 111, 113, 117, 121, 123, 200, 202, 210, 212, 213, 214, 219, 221, 222, 228, 231, 232, 233, 234}

2.2. MIT-BIH preprocessing

The model used in this study receives a series of heartbeats as input. We use Mousavi et al.'s preprocessing of the ECG data to recover individual heartbeats [11]. First, normalize the ECG signal between 0 and 1 interval. The next thing we did was isolate the heartbeat's T-wave place that was connected to the R peak. The continuous signal was then divided into a series of heartbeats by the end of the T-waves, and each heartbeat was annotated in accordance with the MIT-BIH annotation file that was supplied. Ultimately, every heartbeat was resampled to a set length of 280 points. The heartbeat extraction method is straightforward and does not use any techniques for filtering or eliminating noise.

3. Methods

The general flowchart depicted in Fig. 1, which is comprising two elements: (1) the process of the system and its specifics, and (2) the evaluation of model performance. S2SDeepArr consists of two components, the first part is CNN architecture inspired by [19] to extract local and temporal features from the input vector of each heartbeat, and the second one is a sequence-to-sequence network formed by BiLSTM encoder-decoder to integrate the global semantic information formed. At the end of S2SDeepArr, a Dense layer with Softmax is employed for heartbeats classification. The detailed structure is represented in the following Fig. 2.

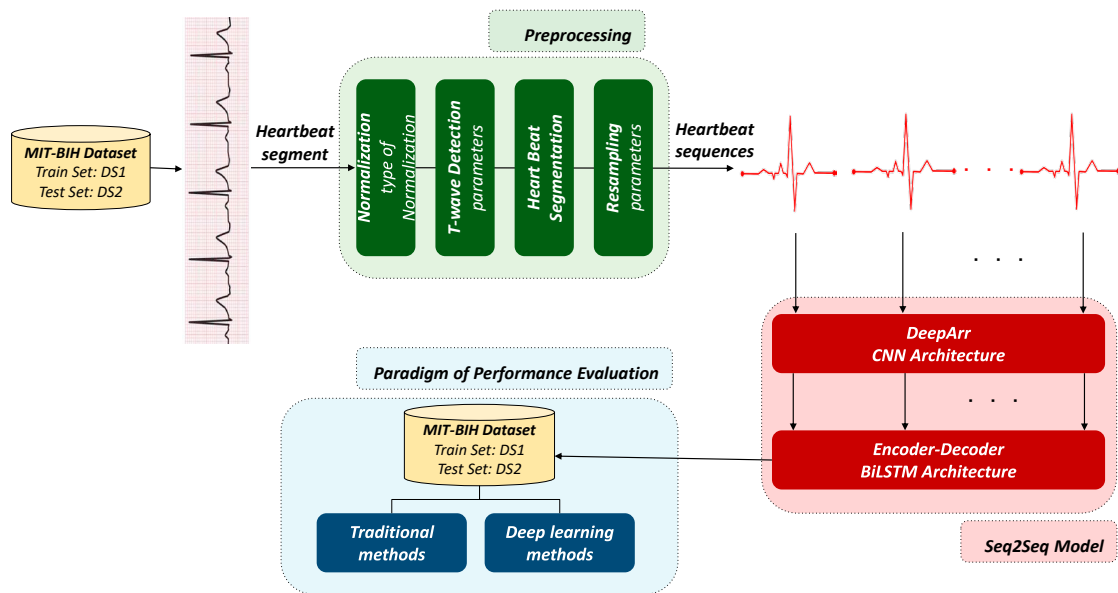


Fig. 1. The overall framework of S2SDeepArr.

3.1. DeepArr CNN

CNN is a widely employed network to extract contextual features representation [23]. CNN primarily comprises convolutional layers to extract features and pooling layers to reduce feature dimension [23]. A deep CNN block

can be constructed by iteratively employing sequences of convolution-pooling layers. As illustrated in Fig. 3, our CNN architecture is organized with three successive convolutional layers and two max-pooling layers. The heartbeat feature is reshaped from 1×280 into 10×28 before being forwarded to DeepArr CNN, then the extracted local features are sent into the Seq2Seq model with a size of $3 \times 1,128$ features. Our proposed CNN plays a role in extracting local features and comprises three blocks, the three convolutional blocks share the same architectural layers but with distinct filter sizes, while consistently employing a kernel size of $K = 3$. The 1D-convolutional layers have 32×2^k filters, where k is initiated at 0 and increments in each convolutional block by 1. Each block consists of four layers and an an

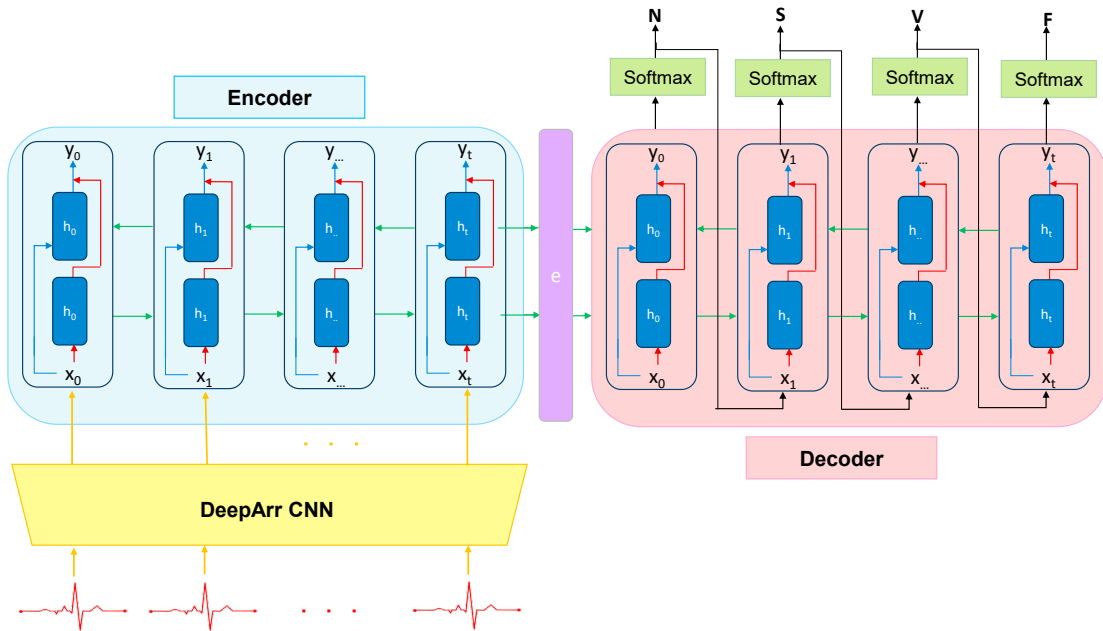


Fig. 2. General architecture of S2SDeepArr.

activation function. First, two consecutive 1D convolutional layers are employed to extract local and temporal feature information from the one-dimensional ECG signal via the sliding window operation in the 1D-convolution layer, and for each convolutional layer, a ReLU activation function is applied. Then, the feature map processed with two 1D convolutional layers is normalized by the batch normalization layer. After that, a Max-Pooling layer of downsampling type with size = 2 and with padding same is added to the first and the second CNN block to reduce twice the dimension of the feature map after the normalization. Furthermore, we implemented a dropout layer with a probability of 0.1 at the end of each convolutional block to make our proposed CNN robust against any overfitting. Feeding the entire heartbeat directly into Seq2Seq would lead to an extended training period and a decrease in performance due to excessive heartbeat sequence length [24]. Therefore, incorporating DeepArr CNN at the beginning of the model enables automatic extraction of the contextual and temporal heartbeat features using two consecutive 1D-convolutional layers, concurrently reducing the sequence length using the Max-Pooling layer. Employing these extracted features, rather than the whole heartbeat, results in a substantial reduction in the duration of training and memory usage.

3.2. Sequence to sequence BiLSTM encoder-decoder network

Global information, including the shape of the heartbeat and its position in the sequence, is a crucial hidden feature. Following the extraction of local features from the ECG, a global information mix must be incorporated after extracting local features from the ECG. We use a sequence to sequence mechanism [25] in the second part of our proposer S2SDeepArr. This mechanism uses an encoder-decoder architecture, which is commonly based on RNN as its fundamental neuron. Nonetheless, RNN is susceptible to issues like short-term memory constraints and the risk of gradient vanishing. For this reason, we adopt the Long Short-Term Memory network (LSTM) [26] as the fundamental neuron.

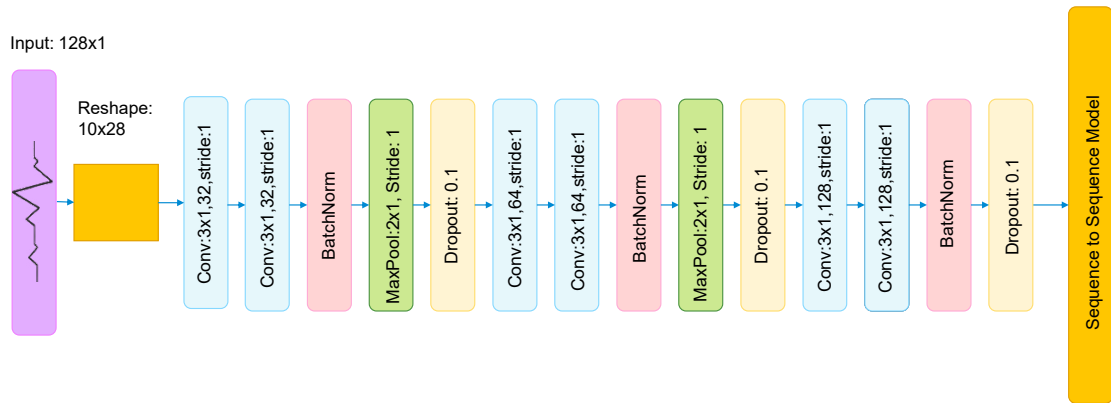


Fig. 3. The proposed architecture.

LSTM, an enhanced version of the recurrent neural network, addresses the limitation of RNN in handling long-distance dependencies. The cell state and the hidden state are the two transition states that LSTM includes, in contrast to RNN's single transition state ht . A structure called a "gate" is used by LSTM to add or remove data from the cell state. LSTM is designed to process data only in one direction the forward. In contrast, the Bidirectional Long-Short Term Memory Network (Bi-LSTM) [27], can update parameters simultaneously in both forward and backward directions. This improves the model's use of context information by allowing it to make use of both past and future data. The parameters of Bi-LSTM are changed through the following process:

$$\begin{aligned}\overleftarrow{ht} &= \tanh(\overleftarrow{W}xt + \overleftarrow{V}ht+1 + \overleftarrow{b}) \\ \overleftarrow{ht} &= \tanh(\overleftarrow{W}xt + \overleftarrow{V}ht+1 + \overleftarrow{b}) \\ yt &= U([\overrightarrow{ht}; \overleftarrow{ht}] + by)\end{aligned}\quad (1)$$

In the BiRNN, $(\overrightarrow{ht}, \overrightarrow{b})$ and $(\overleftarrow{ht}, \overleftarrow{b})$ represent the hidden state and bias of the forward network and backward network respectively. Additionally, xt corresponds to the input of the BiRNN and yt is the output. The target output is generated beat by beat in a sequential fashion using the decoder. The BiLSTM is the fundamental component of the decoder, just like it is the encoder. The decoder obtains a newly generated representation of the input sequence from the encoder to initialize its hidden state. During the initialization of its hidden state, the encoder provides the decoder with a freshly produced sequence representation of the input. Furthermore, the decoder handles the input sequence during processing, wherein it shifts the desired sequence by one position and initiates with a distinctive feature vector called " $\langle GO \rangle$ ". It's important to highlight that the shifted target is employed only pending the training step. Subsequently, an activation function Softmax operation is employed at the end of the decoder to transform the output into a probability vector with a length equal to the number of heartbeat sequences.

The network was trained for 500 epochs with mini-batches of size 20, all the layers used the default initialization method except the initial BiLSTM hidden and cell states were set to zero. All model parameters network weights were updated by the Adam optimizer algorithm with a learning rate was set to 0.0001.

Table 1. Comparison study with existing related works under three categories

Model	ACC	N		S			V			
		SEN	PPV	SPEC	SEN	PPV	SPEC	SEN	PPV	SPEC
S2SDeepArr	99.92	99.91	99.96	99.66	98.86	97.84	99.92	100.00	99.88	99.99
S2SDeepArr*	99.47	99.79	99.38	94.50	84.53	93.26	99.77	99.41	99.88	99.99
Yong Xia [28]	99.90	99.91	99.93	99.37	98.04	97.83	99.92	100.00	99.88	99.99
Zhou et al. [14]	99.83	99.78	99.96	99.62	98.75	94.82	99.79	100.00	99.88	99.99
Jiang et al. [12]	99.89	99.87	99.84	98.56	95.69	99.89	97.06	99.98	99.89	99.97
Mousavi et al. [11]	99.53	99.68	96.05	99.55	88.94	99.72	92.57	99.94	99.50	99.97
Xu et al. [29]	98.50	98.87	98.64	-	83.06	82.48	-	93.46	90.04	-
Niu et al. [30]	96.40	98.90	97.40	-	76.50	76.60	-	85.70	94.10	-
He et al. [31]	95.10	97.50	97.60	-	83.80	59.40	-	80.40	90.20	-
Shi et al.[32]	94.20	95.23	98.92	-	90.74	47.85	-	92.92	84.49	-
Kim et al. [33]	89.10	91.40	97.70	-	49.30	26.60	-	91.40	65.70	-
Ye et al. [34]	86.40	88.50	97.50	-	60.80	52.30	-	81.50	63.10	-

*Without SMOTE method to augment data in the training set.

4. Experimental results

4.1. Performance evaluation

We will illustrate the experimental evaluation in this section. As discussed in the preceding section, the MIT-BIH arrhythmia database has been used to evaluate the proposed S2SDeepArr. To grant a fair comparison with the inter-patient paradigm, we closely followed the protocol proposed by Chazal et al. [22] which is to split the train and the test sets into DS1 and DS2 (detailed in section 2.1). We strictly employed the DS2 only in the test phase and kept it unchanged. The same as that in [11] we test our proposed method on normal beats (N), Supraventricular ectopic beats (S), Ventricular ectopic beats (V), Fusion beats (F), and Unknown beats (Q) classes. To solve the problem of imbalanced data we employed SMOTE [17] to augment the training set into 7000 samples for all categories except the N class, the number of samples for each class is illustrated in Table 1. To evaluate S2SDeepArr performance, four performance metrics are used: positive predictive value (PPV), sensitivity (SEN), specificity (SPEC), and accuracy (ACC). The mathematical formulas are defined below:

$$ACC = \frac{TP + TN}{TP + TN + FP + FN} \quad (2)$$

$$SEN = \frac{TP}{TP + FN} \quad (3)$$

$$SPEC = \frac{TN}{TN + FP} \quad (4)$$

$$PPV = \frac{TP}{TP + FP} \quad (5)$$

In these mathematical formulas, TP refers to true positives, TN represents the true negatives, FP is the false positives, and FN means the false negatives.

4.2. Results

4.2.1. Performance evaluation

To prove the performance of the proposed S2SDeepArr, Table 2 contains a comparison between the relevant performance metrics and the results of related works under the inter-patient paradigm with three classes scenario. To ensure equity, the compared works in Table 2 use the same DS2 dataset, neither more nor less, in the testing phase.

Table 2. Comparison study with existing related works under five categories

Model	ACC	N			S			V			F			Q			
		SEN	PPV		SPEC	SEN	PPV	SPEC	SEN	PPV	SPEC	SEN	PPV	SPEC	SEN	PPV	SPEC
S2SDeepArr	99.81	99.81	99.73		97.81	92.97	96.01	99.85	99.97	99.26	99.95	97.42	95.70	99.97	00.00	nan	100.00
S2SDeepArr*	98.75	99.01	97.70		81.09	48.31	67.04	99.09	99.97	97.57	99.83	60.57	99.16	100.00	00.00	nan	100.00
Yong Xia et al. [28]	97.66	97.35	96.47		71.09	70.26	82.90	99.44	73.92	71.67	96.42	13.40	30.41	99.08	0.00	0.00	99.98
Essa et al. [16]	97.90	98.03	97.48		80.27	65.51	66.19	98.56	93.91	94.55	99.62	19.33	41.67	99.97	-	-	-
Chen et al. [15]	96.77	99.15	97.81		81.96	63.90	76.10	99.23	88.84	96.78	99.80	47.68	54.25	99.68	-	-	-
Wu et al. [35]	96.76	93.15	98.18		86.00	80.23	49.40	96.85	90.99	83.09	98.72	12.63	4.05	97.65	-	-	-
Wang et al. [36]	95.60	91.70	99.10		-	89.90	36.70	-	87.80	96.60	-	55.40	16.30	-	-	-	-
Li et al. [37]	94.61	94.67	99.73		-	27.27	0.16	-	93.74	90.18	-	20.00	0.26	-	-	-	-
Mondéjar-Guerra et al. [2]	94.50	95.90	98.2		-	78.10	49.70	-	94.70	93.90	-	12.40	23.60	-	-	-	-
De Chazal et al. [22]	86.20	87.10	99.20		-	76.00	38.50	-	80.30	86.60	-	89.40	8.60	-	-	-	-

*Without SMOTE method to augment data in the training set.

Our proposed method reaches an average accuracy of 99.92% when the data are balanced with the SMOTE technique for the three classes N, S, and V. On the other hand, in the imbalanced data scenario, we achieved an average accuracy of 99.47%. Across all three classes, S2SDeepArr has achieved the highest SEN, PPV, and SPEC. However, Jiang et al. obtained the PPV value for class S, which is 99.89%, surpassing our PPV for class S, which stands at 97.84%. A comparison with the machine learning approach [34] the proposed method exhibits a 13.52% enhancement in mean accuracy (ACC) without the need for feature extraction procedures, as it utilizes the real ECG heartbeats as input. For deep learning approaches, S2SDeepArr exceeds the proposed method [11] by enhancing the N class's PPV by 3.91% and also for the S class boosting both the SEN and SPEC by 9.92% and 7.35%, respectively and it showed a diminution in PPV metric with -2.05%. Jiang et al. [12] which is use two levels of attention he obtains sub-optimal performance, our proposed model obtains a superior SEN and SPEC of the S class with improvements of 3.17% and 2.86%, respectively. According to the results of Zhou et al. [14], our S2SDeepArr surpasses their capsule model by enhancing the PPV of the S class by 3.02%.

On the other hand, S2SDeepArr shows unsatisfactory results in the imbalanced dataset case (without SMOTE), especially in the minority classes S and V. Compared with the balanced dataset case, in the S class our model decreases the SPEC of the N class by -5.49%. It faces a problem of distinguishing the S class from the N class, which diminishes the SEN also the PPV by -14.33% and -4.58%, respectively in the S class.

Many related works focus on inter-patient arrhythmia classification for four heartbeat classes or even less, aiming to focus on specific types of arrhythmias and simplify complexity features. To guarantee the performance of S2SDeepArr we also tested our model in five-class scenario. Table 3 shows that S2SDeepArr outperforms other approaches in the five-class scenario where our method achieves the best ACC of 99.81% when the heartbeat types are balanced. Compared to the hybrid model proposed in [15], our method enhances all performance metrics in all five classes by improving the average Acc by 3.04%, in the N class the SPEF and PPV are improved by 15.85% and 1.92%, respectively, and in S class the SEN and PPV are also improved by 29.07% and 19.91%, respectively, and in V class the SEN and PPV are also improved by 11.13% and 2.48%, respectively, and in F class the SEN and PPV are also improved by 49.74% and 41.45%, respectively. Also Compared to [28], which utilized a CNN with transformer model, our model improved the PPV and SPEC by 3.26% and 26.72%, respectively in N class, also increasing the SEN and PPV by 22.71%, 13.11%, respectively in S class, for V and F classes the S2SDeepArr is improving the SEN and PPV by 26.05%, 27.59%; 84.02%, 65.29%, respectively. On the other hand, S2SDeepArr shows unsatisfactory results in the imbalanced dataset case (without SMOTE), especially in the minority classes S which decreases the SEN and PPV by -44.65% and -28.97%, respectively, and for F class our model diminishes the SEN by -36.85%. Despite the performance diminution in the imbalanced scenario, our model still have the best results compared with the related work.

To summarize, S2SDeepArr attains the highest overall performance with a 99.29% accuracy in three-class scenarios and a 99.81% accuracy in five-class scenarios. Additionally, its exceptional performance in classes with limited samples and its impressive results in sensitivity, specificity, and positive predictive value metrics demonstrate its ability to reduce the negative impacts of sample unbalance, enabling accurate differentiation in arrhythmia classification.

5. Discussion

To guarantee the performance of S2SDeepArr under the inter-patient paradigm, we compare the results with traditional machine learning and with Deep learning. Table 2 contains a detailed comparison, the machine learning employed in [22, 34, 2] employed a hybrid model, which combines the feature extraction method with classifiers like AdaBoost, random forest, and SVM, and those approaches obtained successful results in arrhythmia classification. Our model is composed of two parts, a DeepArr part and it is CNN blocks to extract features automatically, after the feature extraction process, those features are fed into the second part which is an BiLSTM sequence to sequence model to extract information in two directions. Compared to the traditional machine learning method [5] we reached 3.92%, 5.41%, and 57.94% in mean ACC, N class's SEN, and S class's PPV, respectively.

Many researchers favor deep learning methods for arrhythmia classification, thanks to their capacity for end-to-end heartbeat classification [29, 15, 10, 11, 12, 28, 14]. The efficacy of deep learning architectures was enhanced by the utilization of hybrid models [15, 10, 11, 12, 16, 28, 14, 16] and multiple paths [29]. Nevertheless, class imbalance issues affect negatively deep learning performance, especially with a limited number of samples in minority types. due to this challenge, we use SMOTE to augment the fewer classes only in the training set. The proposed approach reached more than 99% of the performance in the three classes scenario except for the SEN and the PPV of the S class which is more than 97%. Also, in the five-class scenario, our model achieves more than 95% except for the SEN of S class which is obtained a 92.97%.

In the AMII standard of the MIT-BIH dataset, the majority of category samples belong to the N class which reaches 90% of the entire dataset, on the other hand, the remaining heartbeat categories represent less than 10%, which poses a major challenge to the imbalanced class. Deep learning models are sensible to imbalanced class problems, frequently yielding suboptimal outcomes. Similar to [11, 12], we use the SMOTE technique to oversample minority classes. An experimental comparative is employed in Tables 2 and 3 to evaluate the practical effectiveness of the SMOTE method, under the two scenarios, the S2SDeepArr with SMOTE performance is more than without SMOTE, especially for the S category which diminishes the SEN and PPV by -14.33% and -4.58%, respectively. In the five-class scenario, class S decreased the SEN and the PPV by -44.65% and -28.97%, respectively, and for the F class, our model diminishes the SEN by -36.85%.

6. Conclusion

We propose a new arrhythmia classification approach named S2SDeepArr with the inter-patient data and imbalanced data. However, it is important to acknowledge the limitations of our model which faces a problem of distinguishing the S class features from the normal N class in imbalanced cases which is caused by insufficient samples in the train set. Our model offers a wide variety of application prospects as an automatic cardiac rhythm abnormality detection system that may help doctors diagnose patients and assist patients in self-monitoring at any time. In the future, we hope to improve the model's capacity to recognize categories with fewer data and employ a more efficient technique for global information fusion. We use a single-lead ECG instead of the clinically used 12-lead ECG, and we intend to confirm the efficacy of the suggested model on the 12-lead ECG in the future.

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